Support Modules and the False Alarm Budget

Technical Report

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Abstract

Support modules are optional layers that attach to a detection run. They are not models, they have no weights of their own, and one module attaches to any model that declares the input contract it needs. The design problem they solve is narrower than it looks: attaching a second test to a detector calibrated at a false alarm rate q produces a system alerting at roughly $2q$, and every component reports q while it happens. Nothing in the arithmetic complains. This document sets out the taxonomy that makes the false alarm budget divisible, the algebra that divides it, the audit record that says which module caused which alert, and the mathematics of Trial Bed, the first module implemented. Trial Bed is an explainer: it names the pollutant a firing looks like by matching the direction each channel moved against a signature table, and it declines to answer when the channels a signature needs are not measured.

1 Introduction

A detector is calibrated once, against a fault-free distribution, at a false alarm rate chosen in advance. That calibration is the only thing standing between an operator and an alert stream nobody reads, and it is fragile in a specific way: it describes one test. Add a second test beside it and the guarantee is gone, silently, because both tests are individually correct.

This is the failure the support layer is built around. If a primary detector fires with probability q under the null and a second detector fires with probability q under the null, the system fires with probability up to $2q$. Both components report q when asked. No exception is raised, no number looks wrong, and the alarm rate has doubled.

The consequence is that a false alarm budget is a quantity to be divided, never handed out fresh per module. Everything else in this document follows from taking that seriously.

2 Four Kinds

Not everything that attaches to a run costs budget, and collapsing the ones that do with the ones that do not is what makes the budget indivisible. The layer therefore has four kinds, and a module must be exactly one of them. Reading them in order reads the pipeline in order.

A **modulator** moves the alerting threshold and never touches the score. That direction is not a stylistic preference. The threshold was fitted against the fault-free distribution of the score, so a score scaled by

Table 1: The four kinds. Only one of them spends false alarm budget.

Kind	Runs	Method	Changes	Budget
screen	before the model	`admit`	the model's input	none
modulator	alongside	`multipliers`	the threshold	none
detector	after	`score`, `null`	the verdict	yes
explainer	after the decision	`explain`	nothing	none

any factor is no longer a draw from the distribution the fit was made on, and every probability under it quietly stops meaning what it says. Multiplying the threshold instead leaves the score, its probability and the calibration exactly where they were, and turns the adjustment into a stated alerting decision that can be read off the record.

A **detector** adds a score, therefore a test, therefore a second tail to pay for. It is the only kind that spends budget, and separating it from the modulator is precisely what makes the budget divisible. One class with a flag would leave nothing to stop three modules running at q each while the real rate went to $3q$.

A **screen** runs before the model and decides which readings are fit to consume. It adds no test, so it is not a detector and takes no budget; it moves no threshold, so it is not a modulator. What it changes is the model's input, and that carries a consequence neither other kind does: the null and the threshold were fitted on unscreened windows, so screening moves the distribution the score is drawn from and the existing calibration no longer describes it. A screen therefore cannot be switched on without refitting the primary.

An **explainer** runs after the decision has been made and reads it. It adds no test, moves no threshold, and changes nothing the decision was made from. It is a separate kind rather than a detector precisely because charging it budget would cost the primary sensitivity in exchange for a module that can never fire.

3 The Budget

3.1 Dividing q

Let q be the total false alarm rate for the whole run, and let there be n attached detectors with weights $w_1, \ldots, w_n$. The split is

$$q_{\text{primary}} = q\left(1 - \sum_{i=1}^{n} w_i\right), \qquad q_i = q\, w_i. \tag{1}$$

This is a union bound. The probability that at least one of the tests fires under the null is at most the sum of the individual rates, so allocating shares that sum to q puts the combined rate at or below q rather than on it. It is the same correction used whenever a family of tests has to hold a joint error rate [Dunn, 1961], applied here to a family that grows whenever an operator adds a module on the command line.

The default is an even split. With n detectors there are $n+1$ shares, so each weight is $1/(n+1)$ and the sum is $n/(n+1)$, which is below one for every n. Even is the honest default: weighting one detector above another is a claim about which test is more likely to fire, and nothing has measured that.

3.2 Why the weights must sum below one

The weights are required to sum strictly below one, and both boundary cases raise rather than clamp. At a sum of exactly one the primary is left a budget of zero, which is a threshold at infinity and a primary that can never fire. Above one the primary's budget is negative, which is not a rate at all.

Clamping either case would produce a run that looks like it worked, reports a plausible split, and has silently switched the primary detector off. Raising is the only safe behaviour, because the failure is otherwise invisible in exactly the way the whole layer exists to prevent.

Modulators, screens and explainers are absent from this arithmetic entirely. They add no test, so n counts detectors only, and a stack containing nothing but modulators leaves the primary the whole budget.

3.3 Firing on the probability, not the score

Each detector fires when its right-tail probability falls below its own share, $p_i < q_i$, and never on the raw score. Raw scores from different detectors are not comparable, which is the same reason a multi-channel detector converts each channel before combining. Comparing on the probability keeps the budget meaning the same thing for every module regardless of what units its score happens to be in.

A detector's null is fitted at calibration time and loaded from an artifact. It is never fitted here. A detector that fits its null on the window it is judging does not have a null; it has a description of the data it just saw. Where a detector's null is a fitted tail, it is the same peaks-over-threshold construction the primary uses [Siffer et al., 2017, Pickands III, 1975], and where several probabilities have to become one number it is the same combination rule [Fisher, 1925].

4 Composition

4.1 The stack

The modules attached to one run form a stack, built through a loader so that name resolution, contract validation, collision against a model's built-in behaviour, and the budget split all happen before anything runs. A stack that exists is a stack that has already been checked, and a module that cannot attach must not turn up halfway through a detection.

Screens intersect. A reading is admitted only if every screen admits it, and an empty screen list admits everything, which is the identity.

Modulator multipliers compose by product,

$$M_{t,j} = \prod_m M_{t,j}^{(m)}, \qquad M_{t,j}^{(m)} \geq 1, \tag{2}$$

and every multiplier is required to be at least one and finite. This layer raises the alerting bar and never lowers it, so a factor below one is refused rather than applied. A timestep with no reading behind it contributes exactly one, so an uninformed modulator is inert rather than suppressing.

4.2 Ordering

Modules are sorted by name before composition, so the order an operator typed them in cannot reach anything downstream. Sorting rather than trusting commutativity is deliberate. Multiplication commutes, so $ab = ba$ holds, but floating point association does not: $(ab)c$ and $(ac)b$ need not be bit-identical, and three modulators is enough to reach that. An empty stack is the identity with no special case, so zero modules walk the same code as three.

5 The Audit Record

The record answers two questions an operator actually asks: why did this fire, and what changes if I turn that off.

5.1 Three verdicts, not two

Every node at every timestep resolves to one of three verdicts. It fired, it was clear, or it was not scorable. The third exists because a score that could not be computed is not the same answer as a score that did not exceed, and collapsing them is how an unscorable node reaches an operator as a quiet all-clear. A non-finite score never fires, and where nothing was scorable the record says so.

5.2 Counterfactuals

For each attached module the record carries the verdict that would have been reached with that module switched off, computed without re-running anything. A modulator drops out by dividing its multiplier back out of the threshold; a detector drops out by removing its vote. An explainer's entry is written too, and it always equals the actual verdict, which is how the record states in its own contents that the explainer cost nothing.

Table 2: The pollutant signature table as deployed. INDET means the parameter does not discriminate for that pollutant. Entries marked $*$ are expected to move strongly; the current matcher does not yet weight them differently.

	temperature	dissolved oxygen	pH	conductivity	floating cond.
rain	INDET	up	down	down	down
tapwater	down	down	up	down	down
oil	up	down*	down	down	down*
sewage	up	down*	INDET	up	up
fertilizer	flat	down	INDET	up	up

The record extends the primary's own format rather than replacing it. Every field the primary wrote keeps its meaning and the support fields sit alongside, because two formats would mean two readers, and the reader who has to act at two in the morning should not have to know which one they are holding.

6 Trial Bed

Trial Bed is the first module implemented, and it is an explainer. Given a firing it names the pollutant the event looks like, the evidence behind that name, and how much confidence the reporting channels support.

6.1 Directions, not magnitudes

For each parameter the module compares the readings at the timesteps that fired against the readings at the timesteps that did not, and reduces the comparison to a direction. Writing $\bar{b}$ and s_b for the baseline mean and standard deviation and $\bar{e}$ for the event mean,

$$\text{direction} = \begin{cases} \text{UP} & (\bar{e} - \bar{b})/s_b \geq \kappa, \\ \text{DOWN} & (\bar{e} - \bar{b})/s_b \leq -\kappa, \\ \text{FLAT} & \text{otherwise}, \end{cases} \qquad \kappa = 1. \tag{3}$$

Using the baseline's own spread as the yardstick is what lets one rule serve parameters on wildly different scales. A one degree move in temperature and a one degree move in conductivity are not comparable in absolute units; both are comparable in units of how much that parameter normally varies at that node. Where the baseline has no spread at all the comparison falls back to a relative change with a five percent band, and where there is too little data the direction is undefined rather than guessed.

6.2 The signature table

Table 2 is the deployed table. Each cell is the direction a pollutant is expected to push a parameter, and INDET marks a parameter that does not discriminate for that pollutant and is therefore skipped without reward or penalty.

A row is scored by the fraction of its non-INDET parameters whose observed direction agrees,

$$\text{score}(P) = \frac{|\{v \in R(P) : d_v = \sigma_{P,v}\}|}{|R(P)|}, \tag{4}$$

where $R(P)$ is the set of parameters row P asserts a direction for. A row scoring below one half is a poor fit and is reported as a possible new type rather than as a name, as is a tie at the top: if the reporting channels did not separate two candidates, naming one of them would be inventing the separation.

6.3 Declining to answer

The property that matters most is what happens when the channels a row needs are not measured. A row is evaluated only if every parameter it asserts a direction for has an observed direction behind it. Otherwise it is reported as unevaluable, naming the parameters that were absent.

The alternative, scoring a row on whichever of its parameters happen to be present, produces a number that looks exactly like a full match and is not one. A partial signature scored as if it were whole is how a guess reaches an operator wearing the clothes of a diagnosis.

This is not a hypothetical concern for this network. The deployed sites measure conductivity, depth and temperature. Depth is not in the table at all, and every row in Table 2 requires both dissolved oxygen and floating conductivity, with four of the five also requiring pH. On the deployed channels, therefore, *no* row is evaluable, and Trial Bed's honest output at every node is that it cannot evaluate, together with the list of channels that would let it.

6.4 The output

An account names the likely cause, the evidence and the confidence, in a form an operator can act on rather than a label. Where a diagnosis is possible it reads, for example, *consistent with sewage: temperature up, dissolved oxygen down, pH not measured, conductivity up, floating conductivity up.* Where it is not, the same sentence structure carries the refusal and the missing channels. Confidence is reported from how many of the discriminating channels were populated, not from the score, because a perfect match on two channels is not strong evidence.

7 The Modules Not Yet Built

Two further modules are specified and registered but not implemented, and both raise from their required method rather than returning a default. Reaching the raise proves the call arrived.

Newtwork Run is a detector. It will score whether a flagged event travelled the way water does: a real spill appears downstream after the travel lag and a failing sensor does not, so agreement along the flow edges separates the two. Being a detector, it will spend its share of the budget.

Settling Pool is a screen. It will withhold readings a sensor produced while it was settling, purging, or otherwise not measuring the creek, so the model is never asked to explain a reading that was never a measurement. It takes no budget, but as a screen it moves the distribution the primary was calibrated against, so switching it on means recalibrating the primary on screened windows.

8 Limitations and Future Work

The budget split is a union bound, which is conservative. The real combined rate lands below q rather than on it, and how far below depends on how correlated the tests are. Nothing has measured that correlation for this system, so the amount of sensitivity given away is currently unknown.

The even default split is an admission of ignorance rather than a finding. It is the right default only until something measures which test is more likely to fire.

Within Table 2, floating conductivity carries the same direction as conductivity in all five rows. It is nonetheless counted as a discriminating channel, and one discriminating channel is currently enough to unlock a named diagnosis. It also means conductivity's evidence is counted twice in every row's score. Whether that is intended is a question about the table, and the table belongs to the Water Quality team.

No cell in the table carries its own source. The table is attributed as a whole and two cells assert measurable quantities, the local tap water temperature and its pH, with nothing recorded behind them. Depth is measured at every deployed site and appears nowhere in the table, which is part of why no row is currently evaluable.

Finally, Trial Bed reduces a window to one direction per parameter, which discards magnitude and timing entirely. A signature table that distinguished a large move from a small one, or an abrupt one from a gradual one, would need both a richer table and a matcher that reads it, and neither exists yet.

References

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